

1RV24MC042_HARSHAVARDHAN_BR_LAB8

Creating a Kubernetes Pod Using YAML Descriptions

Folder structure

```
nodejs-k8s-pod/  
├─ config.yaml  
├─ pod.yaml  
└─ server.js
```

config.yaml

```
apiVersion: v1  
kind: ConfigMap  
metadata:  
  name: nodejs-app-config  
data:  
  server.js: |  
    const express = require("express");  
    const app = express();  
    app.use(express.json());  
    app.get("/", (req, res) => {  
      res.send("Hello from Node.js and Kubernetes!");  
    });  
    app.listen(3000, () => {  
      console.log("Server running on port 3000");  
    });
```

pod.yaml

```
apiVersion: v1  
kind: Pod  
metadata:  
  name: program-8  
  labels:  
    app: nodejs  
spec:  
  containers:  
    - name: nodejs  
      image: node:18  
      command: ["node", "/usr/src/app/server.js"]  
      volumeMounts:  
        - name: app-code  
          mountPath: /usr/src/app  
      ports:  
        - containerPort: 3000  
  volumes:  
    - name: app-code  
      configMap:  
        name: nodejs-app-config
```

server.js

```
const express = require("express");  
const app = express();  
app.use(express.json());  
app.get("/", (req, res) => {  
  res.send("Hello from Node.js and Kubernetes!");  
});  
app.listen(3000, () => {
```

```
console.log("Server running on port 3000");
});
```

Step 1: Start Minikube and Check Status

```
minikube start
minikube status
```

```
> minikube start
minikube status
🐹 minikube v1.37.0 on Linuxmint 22.1
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
🚧 Pulling base image v0.0.48 ...
🔧 Updating the running docker "minikube" container ...

🚨 Docker is nearly out of disk space, which may cause deployments to fail! (91% of capacity). You c
' to skip this check.
💡 Suggestion:

    Try one or more of the following to free up space on the device:

    1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
    2. Increase the storage allocated to Docker for Desktop by clicking on:
       Docker icon > Preferences > Resources > Disk Image Size
    3. Run "minikube ssh -- docker system prune" if using the Docker container runtime
🍷 Related issue: https://github.com/kubernetes/minikube/issues/9024

🐳 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
   ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
   ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌞 Enabled addons: metrics-server, dashboard, default-storageclass, storage-provisioner
🏡 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 2: Apply Configurations

```
kubectl apply -f config.yaml
kubectl apply -f pod.yaml
```

Step 3: Check Status

```
kubectl get pods
kubectl get deployments
```

```
> kubectl apply -f config.yaml
kubectl apply -f pod.yaml
configmap/nodejs-app-config unchanged
pod/program-8 created
> kubectl get pods
kubectl get deployments
NAME          READY   STATUS             RESTARTS   AGE
program-8     0/1     CrashLoopBackOff   1 (12s ago)  15s
No resources found in default namespace.
```

```
📁 ~/Documents/Obsidian Vault/RVCE_/SEM3/DEVOPS/devopslab/lab8
```

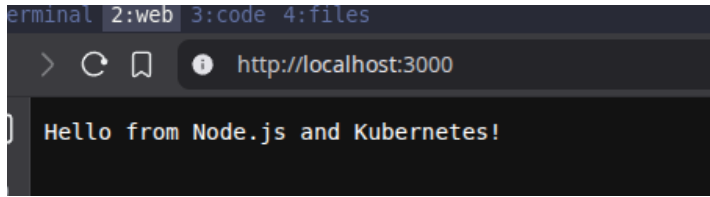
Step 4: Port Forwarding

```
kubectl port-forward pod/program-8 3000:3000
```

```
> kubectl port-forward pod/program-8 4000:4000
Forwarding from 127.0.0.1:4000 -> 4000
Forwarding from [::1]:4000 -> 4000
```

Step 5: Test the Application

```
curl http://localhost:3000
```



```
> curl http://localhost:3000
Hello from Node.js and Kubernetes!
```

Step 6: Clean Up Resources

```
kubectl delete pod program-8 --all
kubectl delete configmap nodejs-app-config
```